Kartik Sharma

Third-year PhD student at ECE, UCLA

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RESEARCH SUMMARY

My work focuses on LLM post-training and agents. I study what a model actually learns at each stage of post-training (SFT, knowledge distillation, GRPO, on-policy self-distillation) and where the useful training signal comes from. I have also built a self-improving multi-agent LLM system that learns from its own errors and defers to humans when uncertain (**NeurIPS 2026**, co-first author), and worked on video-language models for clinical video (**Annals CTN 2026**).

EDUCATION

University of California, Los Angeles California, USA

Ph.D. in Electrical and Computer Engineering. GPA – 4.0

- **Coursework:** Matrix Analysis, Convex Optimization, Linear Programming, Neural Networks & Deep Learning (+ Advanced), Large-Scale Data Mining, Neural Signal Processing.

Indian Institute of Science (IISc) Bangalore, India

Bachelor and Master of Science in Physics, CGPA – 9.0

TECHNICAL SKILLS

Languages: Python, C/C++, Bash
Deep Learning: PyTorch, Hugging Face (Transformers, Datasets, PEFT), vLLM, NumPy, pandas
Methods: RL post-training (GRPO), on-policy (self-)distillation, SFT, LoRA, LLM agents and multi-agent systems, evaluation harnesses, probes, activation steering, calibrated deferral

PUBLICATIONS

1. Shreyas Rajesh*, **Kartik Sharma***, Tonmoy Monsoor, Mehmet Yigit Turali, Richard Idro, Juliana Kayaga, Robert Sebunya, Tracy Tushabe Namata, Jessica N. Pasqua, Vwani Roychowdhury, Rajarshi Mazumder. *Teaching LLMs to Recommend and Defer in Underrepresented Epilepsy Care. arXiv*:2606.31036, 2026. **NeurIPS 2026**.

2. Qiujing Lu*, Tonmoy Monsoor*, Ehsan Ebrahimzadeh, **Kartik Sharma**, Vwani Roychowdhury. *An exterior method for nonnegative matrix factorization.* **ICML 2026**.

3. Lina Zhang, Richard Jiang, Tonmoy Monsoor, Jessica N. Pasqua, Colin M. McCrimmon, Prateik Sinha, **Kartik Sharma**, Muayad Alzuabi, Victor Morales, Hailey M. Miranda, Chaya Manjeshwar, Vwani Roychowdhury, Rajarshi Mazumder. *Video-based automated seizure classification using multimodal large language models. Annals of Clinical and Translational Neurology*, **2026** (accepted).

4. Colin M. McCrimmon, Prateik Sinha, Qing Cao, Tonmoy Monsoor, **Kartik Sharma**, Mehmet Yigit Turali, Ranmal A. Samarasinghe, Vwani Roychowdhury. *Time-resolved neuronal network dynamics distinguish pathological states in organoid models.* **ICASSP 2026**.

5. B. Ananthanarayan, Tanay Pathak and **Kartik Sharma**. *Closed form expressions for certain improper integrals of mathematical physics. The European Physical Journal Special Topics*, Feb 2024.

6. Tushar Pandey, Jason Saroni, Abdullah Kazi, **Kartik Sharma**. *Efficient variational quantum eigensolver methodologies on quantum processors. arXiv*:2407.16107, 2024. Accepted at **QIP 2025**.

* denotes equal contribution.

SCHOLASTIC ACHIEVEMENTS

- **KVPY (SX)** All India Rank **95** (2019); **JEE Main** AIR **376** / ∼1M; **Physics Brawl** Team World Rank **7** (2019); **Mitacs Globalink Research Scholar** (2023, University of Toronto).

TEACHING & MENTORSHIP

- **Teaching Assistant, UCLA:** ECE 131A Probability & Statistics (Fall 2025, Fall 2026); CS M146 Intro to Machine Learning (Spring 2026, incl. guest lecture on LLMs); ECE 232E Large-Scale Networks (Summer 2026).
- **Academic Mentor, RIPS @ IPAM (Analog Devices, Summer 2026):** supervised four undergraduates on cross-modal visual localization (nighttime infrared vs. daytime camera pose).

RESEARCH EXPERIENCE

Auditing the Teaching Signal in On-Policy Self-Distillation Jun 2026

Writeup | *Independent work*

- Reproduced **on-policy self-distillation** (privileged-teacher OPSD) for math reasoning on Qwen3-1.7B/4B; the tuned run improved AIME24 avg@12 from **50.8 to 56.4** and combined pass@12 from 68.3 to 71.7.
- Found two training-pipeline issues (a thinking-token mismatch causing a first-token KL spike, and a KL clip discarding ∼46% of token signal) and ran ablations showing gains depend heavily on these details.
- Built a token-level KL/entropy analysis over 3,000+ traces and hand-labeled teacher corrections: >98% of negative pressure was formatting or phrasing, only ∼0.4% was a reusable reasoning step—per-token KL is not a teaching detector.

Self-Improving Multi-Agent LLMs that Adapt and Defer (NeurIPS 2026) 2025 – May 2026

Under Prof. Vwani Roychowdhury *UCLA*

- Built **Manana**, a multi-agent LLM system that **continually adapts** to each deployment site's practice from expert feedback, through a feedback loop on its own errors and without weight updates (**personalization** without retraining).
- The system mines its own failure logs and **spawns specialized sub-agents** automatically, building task structure and memory with no human in the loop.
- Introduced **Bayesian prompt averaging** over learned prompt states for calibrated confidence and **deferral to human experts**, enabling human–AI collaboration on uncertain cases.
- Evaluated in two real deployment settings with strong distribution shift (US and Uganda hospitals), outperforming monolithic prompting.

Causal Mechanisms of Safety and Refusal in LLMs Sep 2025 – Nov 2025

codebase

- Built an evaluation harness for jailbreak suites (pass@k, refusal rates) to study how **assistant-prefills** causally alter generation.
- Found distinct residual-stream clusters for prefill jailbreaks, and that "tipping tokens" push the model into a refusal state that acts as an **attractor**.

Investigating "Intelligent Slop": Can LLMs Learn From Memes? Dec 2025

Writeup, codebase | *Independent work*

- Tested whether absurdist data with internally consistent logic can induce reasoning gains despite scoring low on conventional data-quality proxies, via **LoRA fine-tuning** of Llama-3.1-8B-Instruct on contrasting subreddits.
- Observed consistent **GSM8K improvements** alongside degradations elsewhere, showing content-dependent capability tradeoffs rather than uniform gains or losses.

Exterior Method for Nonnegative Matrix Factorization (ICML 2026) 2025 – 2026

Collaborator *UCLA*

- Contributed to eNMF, which separates low-rank approximation from nonnegativity: start from the optimal unconstrained factorization, rotate toward the nonnegative orthant, then refine; improves reconstruction error and speed over a large set of NMF baselines.

Video-Language Models for Seizure Classification (Annals CTN 2026) Apr 2025 – Oct 2025

Collaborator *UCLA*

- Applied video–language models (**InternVL**) with pose and optical-flow cues to extract clinical features from seizure video; evaluated against expert annotations.

Bachelor's Thesis (Physics) — Quantum-Enhanced Viterbi Algorithm Dec 2022 – Apr 2023

Under Prof. Shayan Garani *IISc*

- Designed a quantum algorithm accelerating Viterbi decoding for hidden Markov models via a Grover-style closest-string subroutine with **qRAM** access. Thesis